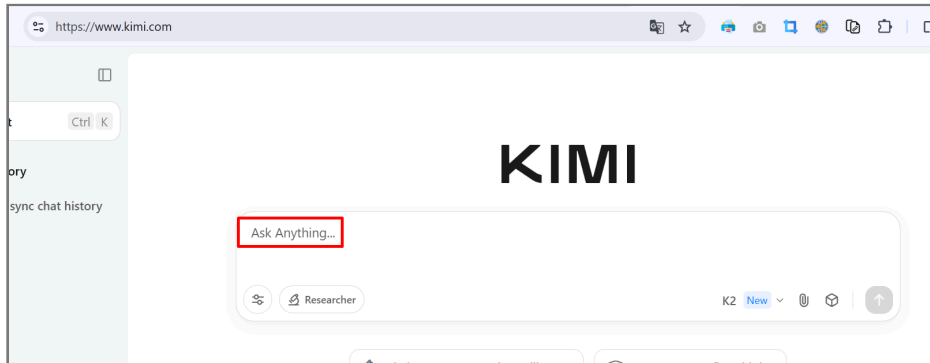


HW2: SQL generation via LLM, and verification via an AI-capable DB :)

Summary: you'll use Kimi K2.6 (a highly capable LLM from Moonshot AI, free to use) to have natural language queries (ie. English questions) be turned into SQL code!

Go to <https://www.kimi.com/chat/>, and sign in using your Google credentials. Now you (and K2.6!) are ready to chat:



Kimi K2.6 excels at code gen and multi-step follow-ups (eg when we ask for each new SQL query in a new prompt): <https://blog.kilocode.ai/p/kimi-k2-first-open-model-to-challenge>. Here's more on Kimi K2.5 (2.6's precursor), from its creators: <https://www.kimi.com/blog/kimi-k2-5.html>

Once you sign in, you can start having K2.6 output SQL, simply by 'chatting' with it, where you would provide:

- a plaintext description of a DB
- followup prompts, one by one, for 5 questions - to have the AI output eqvt. SQL queries

You'll be able to EXECUTE the queries so they run on your tables that have actual data. In other words, the AI makes it easy to 'simply' obtain SQL. During all the decades that SQL has been around, it's been HUMANS alone, who could come up with SQL (eg. look at link after link, to see what this search shows!). But now, for the first time, AI can output SQL too.

Start by copying and pasting this verbatim DB description as your first 'prompt'. This is to provide starting context for the model.

Now that AI has the DB description (from which it can create table schemas), you can continue the chat with it, by asking it to create SQL related to the tables, one question at a time.

Here's what the we want to query - 5 items - input each separately (as its own prompt):

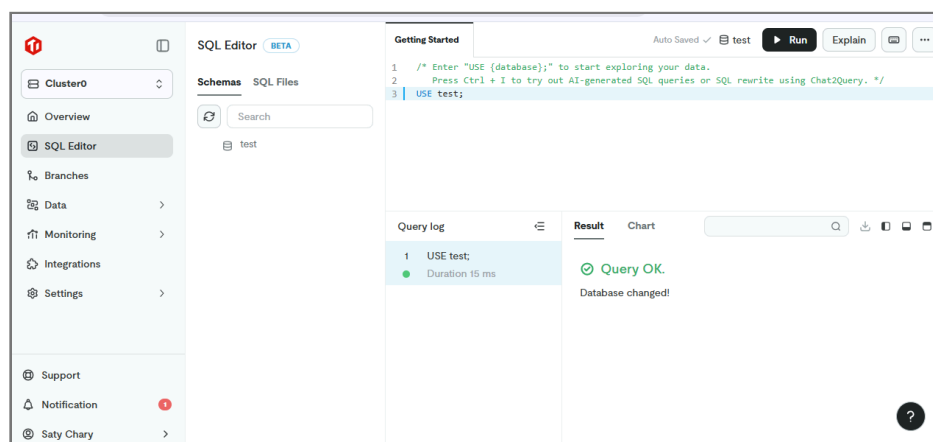
- how much of the \$300,000 loan have we paid off, so far?
- which staff members' licenses will expire next year?
- how many medical procedures did we perform last year?
- can we get a list of insurance companies that our patients are using?
- how many patients are scheduled to visit, during the next 3 months?

For each query, K2.6 outputs any assumptions made, then SQL code!

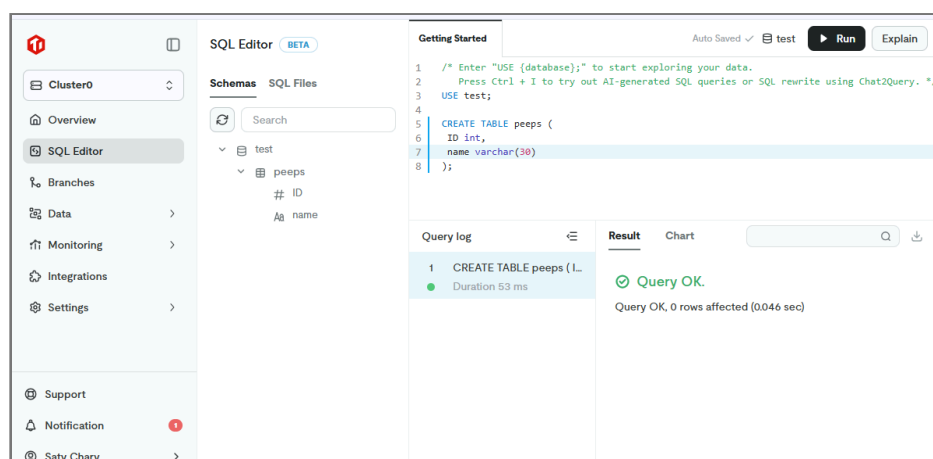
Next we want to execute the queries to see if they work properly. We -could- use K2.6 for this, too, but for practice, let's use a different system :) The idea is to have you get hands-on experience using multiple state-of-the-art RL tools...

Go to <https://auth.tidbcloud.com/> and create a free acct. (eg by signing in with your Google creds) for TiDB, an AI-friendly DB.

With the 'USE' SQL command, create a DB that will hold all your tables (eg. USE HW2;):



After that, you would create the tables, insert some data (eg 10 rows per table) - you can prompt the AI in TiDB to do the [fake ie 'synthetic'] data inserting for you if you like (run the CREATE TABLE commands first, then have the AI do INSERT INTO), or you can insert data 'by hand'. Here is a sample table creation command:



Once the all the data is in, run each query that was output by the Kimi AI, and check (informally) that the output is correct. Take a screenshot of each (query execution, result) for submission.

Please post your qns (and their answers!) on the 'hw2' Piazza page.

What to submit (as a single .zip): for each query, a screenshot showing your prompt to Kimi K2.6 as well as the returned SQL, and your running it in TiDB.

After this course, do play with other systems that generate SQL, eg. <https://github.com/vanna-ai> [you can use this via Streamlit even, eg. see <https://github.com/vanna-ai/vanna-streamlit-simple>]. This explains how Vanna works: <https://github.com/vanna-ai/vanna>. You can also use Qwen 2.5 Coder 32B, Alibaba's model which has been trained using lots of SQL [eg. see pages 19,20 here: <https://arxiv.org/pdf/2409.12186v1>]. You can try deployments at <https://huggingface.co/spaces?q=Qwen2.5-Coder-32B-Instruct>, or even use Ollama to download and run it locally (you do need a beefy GPU, including via a 'DGX Spark' box!!). TL;DR: 'Text-to-SQL' is here to stay! Eg. here, 19 LLMs are compared, re. their SQL gen. capabilities.

HAVE FUN!

